

A Technical Introduction to Hierarchical clustering

Section 1

ALGLIB implements several hierarchical clustering algorithms (single-link, complete-link, Ward) in C++ and C# with $O(n^2)$ memory and $O(n^3)$ run time. ELKI includes multiple hierarchical clustering algorithms, various linkage strategies and also includes the efficient SLINK, CLINK and Anderberg algorithms, flexible cluster extraction from dendrograms and various other cluster analysis algorithms. Julia has an implementation inside the Clustering.jl package. Octave, the GNU analog to MATLAB implements hierarchical clustering in function "linkage". Orange, a data mining software suite, includes hierarchical clustering with interactive dendrogram visualisation. R has built-in functions and packages that provide functions for hierarchical clustering. SciPy implements hierarchical clustering in Python, including the efficient SLINK algorithm. scikit-learn also implements hierarchical clustering in Python. Weka includes hierarchical cluster analysis.

Section 2

In this visualization, each point begins as its own cluster. At each step, the two clusters with the smallest distance between them are merged. On the left, dashed lines indicate which clusters are combined, and a running list of cluster groupings shows how clusters evolve over time. On the right, the dendrogram is constructed simultaneously, with each merge represented by a new branch. The height of each merge in the dendrogram corresponds to the distance or dissimilarity between the clusters being combined. Merges that occur at lower heights represent clusters that are more similar, while merges at higher heights indicate greater dissimilarity. Color-coding is used to associate each merge step in the data visualization with its corresponding branch in the dendrogram.

Section 3

$$d_H(A, B) = \max \{ d_{\sim}(A, B), d_{\sim}(B, A) \}$$

$$\{d_{\sim}(A, B) = \max\{\tilde{d}(A, B), \tilde{d}(B, A)\}\}$$

Section 4

Further reading Kaufman, L.; Rousseeuw, P.J. (1990). Finding Groups in Data: An Introduction to Cluster Analysis (1 ed.). New York: John Wiley. ISBN 0-471-87876-6. Hastie, Trevor; Tibshirani, Robert; Friedman, Jerome (2009). "14.3.12 Hierarchical clustering". The Elements of Statistical Learning (2nd ed.). New York: Springer. pp. 520–8. ISBN 978-0-387-84857-0. Archived from the original (PDF) on 2009-11-10. Retrieved 2009-10-20.

Section 5

Because it disregards vector magnitude, it is frequently used in text mining and document clustering, where documents are represented as high-dimensional term-frequency vectors. Mahalanobis distance accounts for correlations between features by incorporating the inverse of the data's covariance matrix S :

Section 6

Applications in cellular biology HAL-x was motivated by workflows in which biologists need many related clusterings at different granularities (for example to preserve rare leukocyte subsets while still summarizing abundant lineages) without repeatedly training separate models from scratch on millions of cells. The authors report analyses on large cytometry-style datasets where rapid, tunable clusterings are used to support downstream interpretation of cellular phenotypes across conditions.

Section 7

It is often preferred when features are measured on different scales or when the data contain outliers, as it is less sensitive to large deviations than Euclidean distance. Cosine distance measures angular dissimilarity between two non-zero vectors:

Section 8

Complexity The standard algorithm for hierarchical agglomerative clustering (HAC) has a time complexity of $O(n^3)$ and requires $O(n^2)$ memory, which makes it too slow for even medium data sets. However, for some special cases, optimal efficient agglomerative methods (of complexity $O(n^2)$) are known: SLINK for single-linkage and CLINK for complete-linkage clustering. With a heap, the runtime of the general case can be reduced to $O(n^2 \log n)$ instead of $O(n^3)$, at the cost of additional the memory requirements. In many cases, the memory overheads of this approach are too large to make it practically usable. Methods exist which use quadrees that demonstrate $O(n^2)$ total running time with $O(n)$ space. Divisive clustering with an exhaustive search is $O(2^n)$, but it is common to use faster heuristics to choose splits, such as k-means.

Section 9

Overview HAL-x is a hierarchical density clustering algorithm introduced for large, high-dimensional point clouds. Rather than agglomerating solely from a fixed linkage criterion on a precomputed distance matrix, HAL-x builds an initial set of high-density "pure" clusters in a

low-dimensional embedding of a downsampled subset of the data, then merges clusters using supervised linkage: it trains supervised classifiers on the original (unreduced) feature space to measure how reliably two proposed clusters can be separated, and uses those separation scores to decide merge order. The method outputs a hierarchy that can be read as a tree of merges together with a predictive model that can assign labels to held-out or newly collected cells without re-running the entire clustering pipeline on every point during exploratory iterations. In outline, the published implementation proceeds by (i) normalizing and downsampling for dimensionality reduction (commonly t-SNE or UMAP), (ii) estimating density in the embedded space to obtain initial clusters, (iii) constructing a sparse k-nearest neighbor graph between clusters with edge weights derived from classifier-based separability on unreduced coordinates, and (iv) agglomeratively merging the weakest edges while retaining cross-validated accuracy information so users can obtain multiple resolutions from one fitted model.

Section 10

Find the object in this cluster with the most dissimilarity to the rest of the cluster: $i^* = \arg \max_{i \in C_*} \frac{1}{|C_*| - 1} \sum_{j \in C_*} \delta(i, j)$

Section 11

Hausdorff linkage Hausdorff linkage is a hierarchical clustering criterion based on the Hausdorff distance, a metric defined between sets of points in a given metric space. Unlike standard linkage methods that rely on pairwise distances between individual observations, Hausdorff linkage evaluates distances between entire clusters, thereby incorporating information about their geometric structure.

Section 12

Agglomerative: Agglomerative clustering, often referred to as a "bottom-up" approach, begins with each data point as an individual cluster. At each step, the algorithm merges the two most similar clusters based on a chosen distance metric (e.g., Euclidean distance) and linkage criterion (e.g., single-linkage, complete-linkage). This process continues until all data points are combined into a single cluster or a stopping criterion is met. Agglomerative methods are more commonly used due to their simplicity and computational efficiency for small to medium-sized datasets. Divisive: Divisive clustering, known as a "top-down" approach, starts with all data points in a single cluster and recursively splits the cluster into smaller ones. At each step, the algorithm selects a cluster and divides it into two or more subsets, often using a criterion such as maximizing the distance between resulting clusters. Divisive methods are less

common but can be useful when the goal is to identify large, distinct clusters first. In general, the merges and splits are determined in a greedy manner. The results of hierarchical clustering are usually presented in a dendrogram. Hierarchical clustering has the distinct advantage that any valid measure of distance can be used. In fact, the observations themselves are not required: all that is used is a matrix of distances. On the other hand, except for the special case of single-linkage distance, none of the algorithms (except exhaustive search in $O(2^n)$) can be guaranteed to find the optimum solution.